Insights from `Detect_Motion_Starter`

the existing Activity-State baseline (Synheart Kinematics)

Short notes on the starter notebook — what it does, what it gets right, and what it misses. It trains six sklearn models on the 561 precomputed UCI-HAR features; the RBF SVM is its best, reported at 97.93%.

1 The headline accuracy is on an easier 4-class problem

In the cleaning step the three walking types (WALKING, UPSTAIRS, DOWNSTAIRS) are merged into a single MOVING class, and the models are trained on that label. So the reported number is **4-class** (MOVING/SITTING/STANDING/LAYING), not the standard 6-class UCI-HAR benchmark people quote.

Same model (RBF SVM), measured here:

- 6-class (standard): 95.05%
- 4-class (his merge): 97.22%

The merge *removes* the hardest part — telling the walking types apart — instead of solving it. In the 6-class case the errors sit almost entirely in downstairs/upstairs and sitting/standing; laying has 0 errors.

2 No orientation or placement test

The notebook assumes the phone stays on the waist in one orientation, which is how UCI-HAR was recorded. There is no rotation or placement check. For a wearable worn at any angle this is the main gap: rebuilding the same model family on per-axis features and applying a random phone rotation (Fig. 1, right) drops it from 92.2% to 26.6% — a 66-point collapse.

3 The notebook already holds both the robust and the fragile feature

Its own EDA shows the acceleration *magnitude* `tBodyAccMagmean` (orientation-proof) cleanly splits moving vs stationary, while at the same time it uses the gravity-*angle* `angleXgravityMean` (orientation-dependent) to pick out laying. The two are never distinguished — but one survives a rotation and the other does not. That distinction is the whole point of the task.

4 Small bugs

- t-SNE marker list has 6 markers for the 4-class data (errors out).
- Confusion-matrix labels are hardcoded to 6 names for a 4-class model.
- The Gradient-Boosting grid includes invalid 0 values and is ~ 700 fits.

5 Fit to the assignment

561 features is not lightweight or on-device-friendly, and many are orientation-coupled (303 per-axis + 7 gravity-angle). The brief asks for the opposite: few features, lightweight, orientation-proof.

Fair credit: the official train/test split he uses *is* subject-independent (different volunteers in the test set), so his score does generalize across people. The weakness is orientation and the 4-class shortcut, not person-leakage.

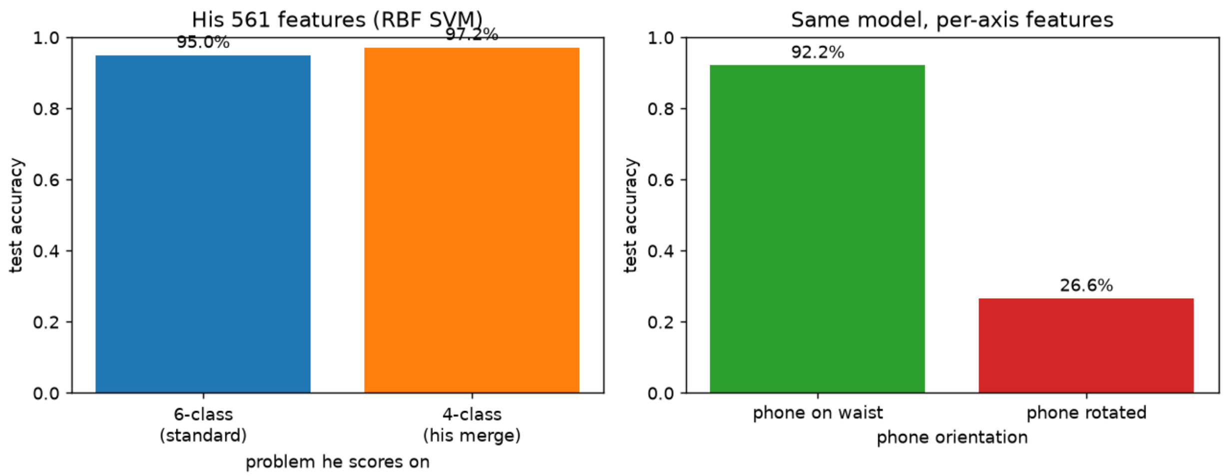

Figure 1: Left: his RBF SVM on the 561 features — the 4-class merge he trains on scores higher than the standard 6-class. Right: the same model family on per-axis features, scored on the waist data vs after a random phone rotation.